Percentages

Quantitative Aptitude | Intermediate | 30 min read

> Master percentage calculations, conversions, and applications. Essential for profit/loss, discounts, and data interpretation.

Quick Reference

Formula	Name	Description
$Percentage = \frac{Part}{Whole} \times 100$	**Basic Percentage**	Converts a fraction or ratio into percentage form. Example: 25 out of 100 = 25%
$Increase\% = \frac{Increase}{Original} \times 100$	**Percentage Increase**	Calculates how much a value has increased. Example: Price from ₹80 to ₹100 = 25% increase
$Decrease\% = \frac{Decrease}{Original} \times 100$	**Percentage Decrease**	Calculates how much a value has decreased. Example: Price from ₹100 to ₹80 = 20% decrease
$Net = a + b + \frac{ab}{100}$	**Successive Change**	Combined effect of two consecutive percentage changes. Example: 10% + 10% = 21%, not 20%
$Fraction \times 100$	**Fraction to Percent**	Multiply fraction by 100 to get percentage. Example: 3/4 × 100 = 75%
$\frac{Percentage}{100}$	**Percent to Fraction**	Divide percentage by 100 to get fraction. Example: 75% = 3/4
$P\% \text{ of } N = \frac{P \times N}{100}$	**Percent of a Number**	Find percentage of any number. Example: 20% of 500 = 100

`N = \frac{P \times 100}{R}`	**Number from Percentage**	If P% of N = R, then N = (R × 100)/P

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Formula Details

Basic Percentage

```
Percentage = \frac{Part}{Whole} \times 100
```

Converts a fraction or ratio into percentage form. Example: 25 out of 100 = 25%

Percentage Increase

```
Increase\% = \frac{Increase}{Original} \times 100
```

Calculates how much a value has increased. Example: Price from ₹80 to ₹100 = 25% increase

Percentage Decrease

```
Decrease\% = \frac{Decrease}{Original} \times 100
```

Calculates how much a value has decreased. Example: Price from ₹100 to ₹80 = 20% decrease

Successive Change

```
Net = a + b + \frac{ab}{100}
```

Combined effect of two consecutive percentage changes. Example: 10% + 10% = 21%, not 20%

Fraction to Percent

```
Fraction \times 100
```

Multiply fraction by 100 to get percentage. Example: 3/4 × 100 = 75%

Percent to Fraction

```
\frac{Percentage}{100}
```

Divide percentage by 100 to get fraction. Example: 75% = 3/4

Percent of a Number

```
P\% \text{ of } N = \frac{P \times N}{100}
```

Find percentage of any number. Example: 20% of 500 = 100

Number from Percentage

```
N = \frac{P \times 100}{R}
```

If P% of N = R, then N = (R × 100)/P

Solved Examples

Example 1: What is 25% of 200?

> **Solution:**
> (25/100) × 200 = 0.25 × 200 = 50

Example 2: If price increases from ₹80 to ₹100, find increase%

> **Solution:**
> Increase = 20, Increase% = (20/80) × 100 = 25%

Example 3: A number is increased by 10% then decreased by 10%. Find net change?

> **Solution:**
> Net = 10 + (-10) + (10×-10/100) = 0 - 1 = -1% (1% decrease) What percentage of 250 is 50?

Example 4: What percentage of 250 is 50?

> **Solution:**
> (50/250) × 100 = 0.2 × 100 = 20%

Example 5: 20% of a number is 40. Find the number.

> **Solution:**
> Let number = x, 0.2x = 40, x = 200

Example 6: A student scored 85 out of 100. What percentage?

> **Solution:**
> (85/100) × 100 = 85%

Example 7: If 30% of x = 150, find x

> **Solution:**
> x = (150 × 100)/30 = 500

Example 8: Price increased from ₹50 to ₹65. Find increase%

> **Solution:**
> Increase = 15, % = (15/50)×100 = 30%

✓ Key Concepts

- ✓ Percentage means "per hundred" or out of 100
- ✓ To convert fraction to percentage, multiply by 100
- ✓ To convert percentage to fraction, divide by 100
- ✓ Percentage points vs percentage change are different
- ✓ Percentage is always calculated on a base value
- ✓ Successive percentages are not additive: a% + b% = a + b + (ab/100)

✗ Common Mistakes

- ✗ Adding percentages directly (10% + 10% = 21%, not 20%)
- ✗ Confusing percentage of what (base value)
- ✗ Wrong base selection in increase/decrease problems
- ✗ Forgetting to multiply/divide by 100 correctly
- ✗ Using percentage points incorrectly

Pro Tips & Shortcuts

💡 Remember fraction equivalents: 1/2=50%, 1/3≈33.33%, 1/4=25%, 1/5=20%, 1/8=12.5%, 1/10=10%, 1/20=5%

💡 For quick mental math: 10% = divide by 10, 20% = divide by 5, 25% = divide by 4, 50% = divide by 2

💡 Always identify the reference value first

💡 Use the formula: Part = (Percentage × Whole)/100

💡 For successive changes, use: Net = a + b + (ab/100)

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